

LAND SECURITY AND MOBILITY FRICTIONS*

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Frictions that impede the mobility of workers across occupations and space are a prominent feature of developing countries. We disentangle the role of insecure property rights from other labor-mobility frictions for the reallocation of labor from agriculture to nonagriculture and from rural to urban areas. We combine rich household and individual-level panel data from China and an equilibrium quantitative framework featuring sorting of workers across locations and occupations. We explicitly model the farming household and the endogenous decisions of who operates the family farm and who potentially migrates, capturing an additional channel of selection in the household. We find that land insecurity has substantial negative effects on agricultural productivity and structural change, raising the share of rural households operating farms by over 40 percentage points and depressing agricultural productivity by more than 20%. Comparatively, these quantitative effects are as large as those from all residual labor-mobility frictions. We measure a sharp reduction in overall labor-mobility barriers over 2004–2018 in the Chinese economy, all accounted for by improved land security, consistent with reforms covering rural land in China during the period. *JEL Codes:* O11, O14, O4, E02, Q1.

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I. INTRODUCTION

The movement of workers from agriculture to nonagriculture and from rural to urban locations is fundamental to the process of structural transformation and economic development. In many developing and transition economies, explicit and implicit frictions impede the mobility of workers across occupations and space. These frictions can have important implications for sectoral and aggregate productivity as well as welfare. Disentangling their effects is key to understanding the drivers of productivity and prioritizing policy initiatives.

China is a country with well-documented migration restrictions that have been the focus of extensive research (Chan and Zhang 1999; Caliendo, Dvorkin, and Parro 2019; Tombe and Zhu 2019). In addition, there are implicit mobility restrictions tied to insecure property rights over farmland (Giles and Mu 2018; Ngai, Pissarides, and Wang 2019; Adamopoulos et al. 2022), frictions that have been identified as important in other developing countries (de Janvry et al. 2015). The risk to farmers of losing their use rights over land if they do not farm it themselves may deter land rentals, work outside agriculture, and migration to urban areas. China offers a unique opportunity to assess the relative importance of alternative mobility frictions that are prevalent in the developing world.

In this article, we combine rich individual- and household-level panel data from China for 2004–2018 and a structural model to quantify the importance of frictions arising from land insecurity and labor-mobility restrictions for agricultural productivity and structural change. For 2004, we find that land insecurity constitutes a substantial barrier to labor mobility that is at least as large in its effect on farm employment and agricultural productivity as all residual labor-mobility frictions. We also find that total barriers to labor reallocation in China fall substantially by 2018, all of which is attributed to improved land security as other labor-mobility costs actually rise slightly. The importance our analysis attaches to improved land security after the early 2000s in facilitating labor mobility is consistent with institutional reforms affecting property rights in farmland, while efforts to relax other labor-mobility barriers have stalled (Chan 2019).

We develop a quantitative framework of structural change and sectoral selection that explicitly models (i) the land rights regime in China and its endogenous effect on labor mobility out

of agriculture; and (ii) the endogenous household decision of who operates the family farm and who migrates, thereby capturing a novel channel of selection in the household. To account for the rich heterogeneity of individual and household choices in the micro data, our model features families and individuals comprising those families. Each individual can work in agriculture in the village as either a farm operator or a wage worker, or migrate to the cities to work in nonagriculture. Individuals may also choose to work part time in farming and nonagriculture. Household members are heterogeneous with respect to their ability in these occupations, which allows for selection across sectors and within the family. We allow for idiosyncratic distortions in household and individual choices.

Following the egalitarian nature of the land tenure arrangement in China, we assume that all families are allocated the same use rights over farmland. Farmers can adjust the scale of their operation through land rentals. We capture land insecurity in the model in a tractable way through (i) the risk that families lose their use rights to land they rent out, and (ii) a family-specific perceived income loss associated with the loss of this land. In our setup, land insecurity not only deters farmers from renting out land along the intensive margin but also deters them from completely abandoning the land in the village out of fear of losing access entirely, thus acting as a *de facto* barrier to labor mobility out of agriculture.

In addition, individual labor supply choices are subject to residual idiosyncratic occupational labor-mobility frictions, a catchall for all other institutions or barriers, such as an individual's *hukou* and explicit migration costs, which may impede the mobility of workers out of agriculture and affect sorting. These individual-level mobility frictions are distinct from the perceived land insecurity at the household level, which emanates directly from the household's participation in the land rental market.

Our framework gives rise to rich patterns of selection within households and across occupations that have implications for agricultural and aggregate productivity and structural change. We quantitatively evaluate the role of land insecurity and other labor-mobility frictions for these outcomes, exploiting our micro-level data on individuals and households for China between 2004 and 2018. The data provide detailed information on labor supply, farm production, and incomes by occupation to which we calibrate idiosyncratic abilities and distortions in our model.

In particular, the model allows us to identify separately land insecurity from labor-mobility frictions by exploiting the fact that land insecurity is primarily a family-level friction, manifesting through the family decision of whether to operate a farm, whereas the labor-mobility barrier operates at the individual level. As a result, land insecurity is pinned down empirically from the share of village families operating farms, while labor-mobility barriers are linked to the sectoral income gap between agriculture and nonagriculture. We directly measure the risk of land loss tied to land rental from supplementary survey data we collected in China, expressly for the purpose of measuring land insecurity. We draw on data on the observed historical frequency of village-wide land reallocations and households' responses to the perceived likelihood that they lose access to the land in such an event.

We find that in 2004 labor-mobility frictions associated with land insecurity are high in levels and large in dispersion and responsible for an inefficiently large number of farms in agricultural production. Moreover, the effects of land insecurity on employment and agricultural labor productivity are at least as large in magnitude as the residual labor-mobility frictions. From the perspective of 2004, introducing secure land rights substantially reduces the share of households operating farms from 74% to 28% and lowers the share of employment in agriculture among village households from 56% to 46%, resulting in an improvement in agricultural productivity of 25%. By comparison, while removing residual labor-mobility frictions has a similar effect on the share of labor in agriculture, it only reduces the percentage of village households farming to 49% and increases agricultural productivity by 16%.

Over 2004–2018 we find a marked improvement in land security but a slight increase in other labor-mobility frictions. Our analysis highlights that much of the increase in labor mobility over the period is tied to improvements in land tenure security, consistent with the measured decline in the frequency of village-wide land reallocations associated with policy reforms pursued in China since the early 2000s. We also find an important interaction between land insecurity and misallocation in agriculture as the productivity gain from an efficient reallocation of resources in agriculture more than doubles when land is secure. These additional gains arise from improved selection in farming in households and improved selection across sectors.

We extend our main framework along three important dimensions. First, we consider a local nonagricultural sector in the rural area, motivated by the striking regularity in the micro data for China that among those with rural *hukou* working in nonagriculture, more than half are employed locally, with the rest migrating to the cities to work. Adding this local dimension allows the framework to distinguish between sectoral and spatial reallocation. Second, we also consider age differences among family members, motivated by the empirical regularity that farms are typically operated by older people in the household, with younger members more likely to seek employment opportunities outside farming. In 2004, individuals over the age of 45 represented 40.5% of the labor force in the countryside, but were operators of 57.8% of all farms. Third, we examine regional heterogeneity in the extent of land insecurity and labor-mobility barriers, motivated by geographic differences in proximity to urban areas.

Overall, we find that our conclusions about the role of land insecurity and other labor-mobility frictions for aggregate outcomes are robust to these extensions. The extended framework offers richer micro-level implications. For those with rural *hukou*, we find it is much less costly to work part-time in rural nonagriculture than in urban nonagriculture as rural nonagriculture does not involve spatial reallocation. Nevertheless, labor mobility barriers are actually higher for the rural nonagricultural sector, which can reflect higher nonpecuniary benefits in cities or larger labor market imperfections in rural areas. We also find that older workers face substantially higher overall barriers to labor mobility compared with younger workers, which helps explain why they are more likely to operate farms. Our analysis also suggests important village-level differences in labor mobility in China, with peri-urban areas facing less severe land insecurity and lower local labor-mobility barriers than more remote areas.

This article relates to several strands of literature. In highlighting the role of farming for labor supply choices, our work relates to the literature on structural transformation and agriculture (Gollin, Parente, and Rogerson 2002; Restuccia, Yang, and Zhu 2008) and the agricultural productivity gap (Gollin, Lagakos, and Waugh 2014a). The article is also connected to the agricultural productivity literature emphasizing misallocation (Adamopoulos and Restuccia 2014), particularly that relating to land market institutions (Chen 2017;

Gottlieb and Grobovšek 2019; Adamopoulos and Restuccia 2020; Chari et al. 2021; Adamopoulos et al. 2022), and sectoral selection (Lagakos and Waugh 2013; Alvarez 2020; Hamory et al. 2021; Adamopoulos et al. 2022). Equally important is the literature on migration costs and structural transformation (Bryan and Morten 2019; Lagakos et al. 2020; Schoellman 2020; Hamory et al. 2021; Morten and Oliveira 2024).¹ While these papers focus on estimating the magnitude of migration costs using reduced-form or structural approaches, we aim to disentangle the implicit migration cost arising from insecure land rights from other labor-mobility barriers. Our study is also broadly related to the literature on growth and development in China (Song, Storesletten, and Zilibotti 2011; Brandt, Tombe, and Zhu 2013; Brandt, Kam-bourov, and Storesletten 2020, among others) and the literature on institutions as an obstacle to development in poor countries (Acemoglu, Johnson, and Robinson 2001, 2005).

The two papers most closely related to ours are Ngai, Pissarides, and Wang (2019) and Adamopoulos et al. (2022). Ngai, Pissarides, and Wang (2019) highlight that land insecurity can manifest as a labor-mobility barrier out of agriculture in China. The value added of our article is a quantitative assessment of the effect of land insecurity versus a catchall of other labor-mobility barriers in a rich, empirically grounded model that nests individual and family selection. The rare individual and household micro-level data we have for China facilitate this investigation, the identification of the two types of barriers, and the quantification of selection. Quantitatively, selection is the largest component of these effects. Adamopoulos et al. (2022) highlight the effect of China's land institutions for misallocation within villages in agriculture and sectoral selection using household-level data for an earlier period. In contrast, we use individual- and household-level data to estimate idiosyncratic mobility barriers and study labor allocation and migration decisions in households, thus allowing us to examine misallocation in households and

1. Previous work has studied several factors that could deter labor mobility, including transportation infrastructure (Asher and Novosad 2020), land use rights (Field 2007; de Janvry et al. 2015), rural insurance networks (Munshi and Rosenzweig 2016), monetary cost and risk (Bryan, Chowdhury, and Mobarak 2014), administrative registration and services (Chan and Zhang 1999), housing (Brueckner and Lall 2015), and regional variation in industrialization (Eckert and Peters 2023).

across space. Relative to the previous literature, we also model differences between young and old adults in their ability and mobility barriers to capture structural transformation by age (Hobijn, Schoellman, and Vindas Quesada 2018; Porzio, Rossi, and Santangelo 2022).

The article proceeds as follows. The next section describes the land and labor-mobility institutions in China. Section III documents key regularities of China's structural transformation that motivate our framework of analysis. In Section IV, we present the main model, and Section V estimates the model by matching moments from micro and aggregate data for 2004. Section VI performs quantitative experiments to assess the relevance of land security and other mobility frictions and their evolution over time. We also evaluate the interaction of land insecurity and farm-level distortions associated with China's land institutions. In Section VII, we extend the main framework by adding spatial choices in nonagricultural employment, age differences, and regional village heterogeneity. We conclude in Section VIII.

II. INSTITUTIONAL BACKGROUND

We provide institutional context for land and labor-mobility policies in China since our framework aims to decompose the costs associated with out-migration of workers from agriculture into those related to land market institutions and other labor-mobility frictions.

II.A. Land Market Institutions

With the introduction of the Household Responsibility System in the early 1980s, rural households were granted use rights and residual income rights over land. Farmland was not privatized, however, and ownership continued to reside in the collective, that is, the village. The allocation of village land was highly egalitarian and tied to "membership" in the village through the household registration system. As a result, per capita differences between households in landholdings within a village were small.

Initially, use rights were extended to households for 15 years. Through the late 1990s, however, local officials often carried out village-wide reallocations of land across households to accommodate demographic changes (Benjamin and Brandt 2002). Rent-seeking behavior on the part of village leaders was also likely a

factor (Kung and Liu 1997; Brandt et al. 2002). In addition, rural land was occasionally expropriated from households by the state for nonagricultural uses, including highway construction, urban development, and industrial parks.

In most villages, land rental was not officially restricted, but rental markets were thin, with less than 3% of total land rented out by the mid-1990s (Brandt et al. 2002). “Use-it-or-lose-it rules” were likely partially responsible. Village leaders often regarded rental transactions as a signal of land misallocation and viewed village land reallocations as an opportunity to redirect farmland to other households. In the case of land expropriation by the state, households who rented land out risked not receiving compensation. Since rentals invited dispossession, households were unwilling to rent out their land.

This incomplete set of property rights over land had two important implications. First, land was misallocated within villages, a product of the highly egalitarian distribution of use rights and limited market-based exchange through rental (Adamopoulos et al. 2022). Second, fear of the loss of land use rights (or compensation in the event of expropriation by the state) prompted households to assign family members to farm the land to protect their rights.

A series of reforms to land policy in China beginning in the late 1990s helped strengthen household property rights but did not privatize farmland. In 1998, the Land Management Law extended the use rights to rural households for 30 years. In 2003, the Rural Land Contracting Law codified legal rights of rural households for leasing out agricultural land. As a consequence of these reforms, village-wide reallocations fell considerably, as we document in Online Appendix A.5. Estimates suggest that these reforms led to a 7% increase in land rentals, and the reallocation of land toward higher-productivity farm households (Chari et al. 2021). In 2013, further reform of land policy allowed farmers to transfer their land use rights to others, including large commercial farms. In 2014, the right to mortgage the use of land was extended to rural households (Zhou, Liang, and Fuller 2021). Finally, in 2018, a formal land titling/certification process was begun that by 2021 was largely complete.

II.B. Labor-Mobility Institutions

Labor mobility in China has been heavily influenced by the household registration or *hukou* system (Chan 2019). Established

in the 1950s, the primary purpose of the system was to control migration between the countryside and the cities. Under the *hukou* system, each individual is assigned agricultural or nonagricultural *hukou* in a locality, for example, village, town, or city, which determines where they can live and work. It also determines their rights and access to local public services. Major differences exist between rural and urban areas in the level and availability of these services, as well as between cities since larger cities typically confer much better benefits than smaller cities.

Through the late 1970s, there were very tight restrictions on migration and on the type of nonagricultural activities that were permitted in the countryside. Restrictions on the type of activity began to be relaxed in the early 1980s, followed by an easing of restrictions on rural *hukou* holders from moving to the cities to work (de Brauw and Giles 2018). Rural migrants continued to be restricted in terms of the public services they could access once they moved. Data from China's population census reveal that migrants' share of nonagricultural employment rose from 6.4% in 1990 to 17.7% in 2000, and then to 21.0% in 2005.

In the early 2000s, new reforms made it easier for migrant workers to obtain working permits in other provinces, with some coastal provinces totally eliminating the requirement (Sun, Bai, and Xie 2011; Kinnan, Wang, and Wang 2018). In 2014, the distinction between agricultural and nonagricultural *hukou* in the same location was eliminated, thereby entitling, on paper, all residents to the same set of local public services. Chan (2019) argues that this reform was primarily limited to smaller cities and that it actually became harder to obtain local *hukou* in larger coastal cities, which restricted in-migration and even tried to force migrants out by limiting migrants' children's access to education. Rising housing prices in the cities likely reinforced the effects of weak enforcement of reforms and contributed to rising migration costs for rural families.

Data from the Population Census for 2005, 2010, and 2015 reveal a slowing in labor flows, with migrants' share of nonagricultural employment rising to 26.5% in 2010, but to only 28.3% by 2015. Consistent with these trends, Wu and You (2023) find that the fraction of migrants obtaining local *hukou* also experienced a pronounced reduction over the same period. For individuals who had completed high school, for example, the fraction receiving urban *hukou* within five years fell from 16.3% and 38.6% in tier 1 and 2 cities in 2000, respectively, to only 3.9% and 10.5% in 2010.

III. STYLIZED FACTS ON CHINA'S STRUCTURAL TRANSFORMATION

The Chinese economy has experienced substantial structural change that has been accompanied by a marked reduction in the share of the labor force engaged in agriculture. Between 1995 and 2018, the share dropped from 48.0% to 13.7%. This decline reflects two forces: the increase in the percentage of households living and working in the cities and changes in the employment patterns of households registered in the countryside.

We draw on the nationally representative rural household survey data collected by the Research Center for the Rural Economy (RCRE) under the Ministry of Agriculture of China to describe salient features of the changes in the countryside that motivate our framework and analysis. For a detailed description of the earlier data, see [Benjamin, Brandt, and Giles \(2005\)](#). We focus on the period between 2004 and 2018, leveraging the individual-level labor supply data that were added to the survey in 2003. Our unbalanced panel contains information on over 20,000 households a year drawn from 300 villages throughout China. [Online Appendix A](#) provides a detailed description of the data.

III.A. Agricultural Employment

The first row of [Table I](#) reports the falling share of China's total labor force employed in agriculture between 2004 and 2018. This decline reflects two forces: The reduction in the share of households registered in rural areas, which fell from 69.1% in 2004 to 39.1% in 2018, and the reallocation of labor from agriculture to nonagriculture among rural residents.² Among those registered in the countryside, the share working in agriculture fell from 69.8% to 44.2%. The largest reductions occurred among full-time farm operators and part-time workers, which fell 10.9 and 8.7 percentage points, respectively. Other household members working full-time in agriculture fell 6 percentage points. A simple decomposition suggests that the reallocation of labor among rural residents from agriculture to nonagriculture was the source of half of the total reduction in the nationwide share of employment in agriculture.

2. In each year, the share of nationwide employment in agriculture is the product of the share of households living in rural areas and the share of labor among these households in agriculture.

TABLE I
EMPLOYMENT IN AGRICULTURE

Variable	2004	2009	2013	2018
Nationwide agricultural employment share (%)	39.1	28.4	20.9	13.7
Share of all households living in rural area (%)	69.1	57.2	46.4	39.1
Share of labor days in agriculture among rural households (%)	56.5	49.6	45.1	35.1
Share of individuals in rural areas involved in agriculture (%)	69.8	62.0	56.7	44.2
Share of farm operators (%)	27.6	23.8	21.9	16.7
Share of full-time workers (%)	21.5	20.3	18.7	15.5
Share of part-time workers (%)	20.7	17.9	16.1	12.0
Share of individuals in rural areas in full-time nonagriculture (%)	30.2	38.0	43.3	55.8

Notes. Nationwide statistics adjusted from the Chinese Statistics Yearbook. All other statistics are calculated from micro-level data from the RCRE survey, as shares out of the total numbers of employed individuals in the survey. For each year the shares of farm operators, full-time, and part-time workers sum up to the total share in agriculture, save for rounding.

TABLE II
AVERAGE FARM SIZE AND LAND RENTALS

Variable	2004	2009	2013	2018
Average farm size (ha)	0.59	0.62	0.71	0.91
Share of households renting in land (%)	17.9	20.0	17.9	18.7
Share of land rented in (%)	14.4	22.4	18.6	31.8

Notes. Average farm size is the total cultivated land over the total number of operating farms (households) from the RCRE survey data and expressed in hectares (ha). Statistics on land rentals are calculated among farming households in the survey each year. Data are from authors' calculation from the micro-data sample.

In 2004, 50.9% of rural people worked in nonagriculture, of which 30.2% were full-time, and 20.7% were part-time (see the last two rows of Table I). Between 2004 and 2018, the percentage of people working in nonagriculture increased from 50.9% to 67.8% of the rural labor force. All of this increase is a product of the growth in full-time employment in nonagriculture, which grew from 30.2% to 55.8%. The share working part-time fell from 20.7% to 12.0%.

III.B. Average Farm Size and Land Rentals

In Table II we report for select years average farm size of farming households in the RCRE survey, measured as total

TABLE III
STRUCTURAL TRANSFORMATION BEFORE 2004

Variable	1995	2002
Nationwide agricultural employment share (%)	48.0	42.2
Share of households living in rural area (%)	75.4	72.6
Share of labor days in agriculture among rural households (%)	63.6	58.2
Share of individuals in rural areas involved in agriculture (%)	73.6	67.5
Share of full-time workers (%)	53.6	48.8
Share of part-time workers (%)	19.9	18.8
Average farm size (ha)	0.57	0.59

Notes. Nationwide statistics adjusted from the Chinese Statistics Yearbook. Employment statistics are calculated from micro-level data from the RCRE survey for 1995 and 2002, as shares out of the total numbers of employed individuals in the survey. Average farm size is the total amount of farm land over the total number of operating farms (households). The numbers are not directly comparable to those in the survey over 2004–2018.

cultivated area divided by the number of farming households. Average farm size was only 0.59 hectares (ha) in 2004, and increased by more than 50% by 2018, but still remained below 1 ha.

Underlying the increase in average farm size is an increase in rentals of farm land. In [Table II](#) we also report the share of farming households that rent in land (second row), which can be thought of as the extensive margin of renting, and the share of land rented in among farming households, or the intensive margin. Although the share of rural households renting in land increased slightly, the share of land rented in increased more, from 14.4% in 2004 to 31.8% by 2018. Hence, the expansion in land rental is primarily driven by the intensive margin and increases in the average amount of land rented in by households in the land rental market. The increase in land market activity is consistent with the alleviation of frictions in land markets through reforms that improved land security in China (see, for example, [Chari et al. 2021](#)).

III.C. Earlier Period 1995–2002

We draw on earlier waves of the RCRE survey to put structural change in the Chinese economy between 2004 and 2018 into perspective. Before 2004, there are signs of structural change, however, the process was much slower (see [Table III](#)). Between 1995 and 2002, for example, the share of employment in agriculture at the national level fell from 48.0% to 42.2%, or one-half the annual rate that we observe afterwards. This reflects only modest changes in the share of households living in rural

areas and the allocation of labor by these households to agriculture. Average farm size also remained more or less constant.

These features of China's structural transformation, which are prevalent in other lower-income and emerging economies, motivate the framework we develop in the next section.

IV. A MODEL OF FRICTIONAL SELECTION

To study the role of insecure property rights and other labor-mobility frictions on reallocation and aggregate outcomes, we develop a general equilibrium model of occupational and sectoral choices of heterogeneous individuals subject to idiosyncratic distortions. Our Roy-type model (Roy 1951) of occupational choice has four key novelties. First, we consider families that make an endogenous decision of who in the family operates the family farm. Second, we introduce family-level insecure property rights over rented farmland. Third, we consider residual labor-mobility barriers to capture all factors other than land insecurity that can impede individual labor mobility. Fourth, in contrast to the standard Roy model, where occupational choices are binary, we allow for part-time employment. We extend this framework in Section VII to allow for spatial choices in nonagricultural employment, age differences in families, and regional village heterogeneity.

IV.A. Description

The population is organized into families living in villages and urban centers. At each date, two goods are produced: an agricultural good (a) and a nonagricultural good (n). In our main specification agricultural production is limited to villages while production of the nonagricultural good occurs in urban centres. Later, we extend the model to include rural nonagricultural production. Village households can work locally in agriculture or migrate to the cities to work in the nonagricultural sector. Urban households work in the nonagricultural sector. We focus on the factors influencing the choice of occupations and locations of individuals and families living in villages.

1. *Preferences and Endowments.* All households have preferences over agricultural and nonagricultural goods represented

by the non-homothetic utility function:

$$(1) \quad u(c_a, c_n) = \phi \log(c_a - \bar{a}) + (1 - \phi) \log(c_n), \quad \phi \in (0, 1),$$

where c_a is consumption of the agricultural good, $\bar{a}$ is a minimum (subsistence) consumption requirement of the agricultural good, and c_n is consumption of the nonagricultural good. This specification of preferences allows us to capture income effects as a source of structural change. Our results are robust to using alternative specifications, such as the price independent generalized linear preferences (Boppart 2014), with results reported in [Online Appendix G](#).

There are two types of households: village and urban. Urban families are homogeneous and make no production decisions. Each is endowed with one unit of labor of average ability $\bar{h}$ and is engaged in nonagricultural production. There is an exogenous mass N_n of urban families.

Village families are at the core of all decisions relating to agricultural production, occupational choice, and migration to the city. Village families can be engaged either in agriculture locally or in nonagriculture in the city. There is a unit measure of village families indexed by i , comprising J individuals per family, indexed by $j = 1, 2, \dots, J$. Each family member is endowed with a pair of abilities (s_{ij}, h_{ij}) , where s_{ij} is ability as a farm operator and h_{ij} is earning ability in nonagriculture. Families are subject to insecure property rights over rented farmland, while individual family members are subject to idiosyncratic labor-mobility frictions. Individual-level abilities and idiosyncratic frictions are drawn from a joint multivariate distribution with a cumulative distribution function Φ .

Since there is a unit measure of village families, the total number of individuals in villages is J , and the total amount of farm land is L , which we normalize to one without loss of generality. Use rights over farmland are allocated across all village families on an egalitarian basis, with each family endowed with an amount of land $\bar{\ell}$. Individuals are endowed with one unit of time in each period that is supplied inelastically and can work as a full-time farm operator or as an agricultural or nonagricultural worker on a full-time or part-time basis.

2. *Technologies.* The nonagricultural good is produced using a constant returns to scale technology,

$$(2) \quad Y_n = A_n H_n,$$

where A_n is the productivity parameter of the nonagricultural sector and H_n is the total labor input in efficiency units.

The agricultural good is produced by family farms. A farm is a decreasing-returns technology that requires the inputs of a farm operator with managerial ability s_{ij} , cultivated land l_{ij} , and farm labor n_{ij} :

$$y_{ij} = A_a s_{ij} \left(l_{ij}^\theta n_{ij}^{1-\theta} \right)^\gamma,$$

where y_{ij} is agricultural output, A_a is an agricultural technology parameter, $\theta \in (0, 1)$ captures the importance of land relative to labor in production, and $0 < \gamma < 1$ captures returns to scale at the farm level. Note that farm productivity is determined by the ability of the family member that is chosen to operate the farm s_{ij} , which in turn governs the input choices of the farm.

3. *Land Rental Market.* There is a potentially active market for rentals of farmland that allows a farmer to adjust the scale of their farm operation relative to their endowed use rights $\bar{\ell}$. Their land input is then $\ell_{ij} = \bar{\ell} + \ell_{ij}^{\text{rent}}$, with $\ell_{ij}^{\text{rent}} > 0$ for farmers that rent in land and $\ell_{ij}^{\text{rent}} < 0$ for those that rent out land. Let q denote the relative rental price of land in the market.

4. *Other Markets.* A representative firm operates the nonagricultural technology in a competitive market. We set the nonagricultural good as the numeraire and normalize its price to one. We denote by w_n the competitive price of efficiency units of labor in the nonagricultural sector. Profit maximization implies that the wage equates the marginal product of efficiency units of labor,

$$w_n = A_n.$$

In addition to land, farms operate in competitive markets of labor and output, subject to frictions that we describe below. We denote by p_a the relative price of the agricultural good and by w_a the wage rate in agriculture, which can differ from the wage rate in the nonagricultural sector.

IV.B. *Idiosyncratic Distortions*

Families and individuals are subject to idiosyncratic land and labor-mobility distortions.

1. *Insecure Land Rights (Family Level).* We explicitly model the insecurity in property rights over farmland that families in China face. In principle, land rentals are not prohibited. However, families that rent out land risk losing access to this land in the event of an administrative land reallocation. We denote by η the probability that households lose the use rights to land they rent out, without compensation. These households suffer a family-specific income loss equal to φ_i per unit of land rented out. Differences in φ_i across families reflect differences in the valuation of the land tied to observable and unobservable characteristics of the land and farming family. Overall, the product $\eta\varphi_i$ captures the family's perceived income loss associated with land insecurity from renting out land. We assume that in the event of an administrative land reallocation, it is realized after land input decisions have been made and after production has taken place.

2. *Labor-Mobility Frictions (Individual Level).* Individual labor supply choices outside agriculture are subject to idiosyncratic distortions, modeled as barriers on wage income. An individual j in family i faces a wage barrier ξ_{ij} if they work in the nonagricultural sector. This barrier is a catchall for all the factors that may impede the reallocation of labor to the nonagricultural sector, exclusive of the land institutions. Note that the land market frictions described above are linked directly to a family's participation in the land rental market, while the labor-mobility friction is linked directly to migration to the city.

3. *Residual Distortions in Agriculture.* Following Adamopoulos et al. (2022), we assume that operating a family farm is subject to an implicit distortion on output τ_{ij} imposed on individual j in family i . This distortion encompasses the residual frictions in agricultural production that cannot be directly captured by the land loss risk described above and has the property that more productive individuals cannot necessarily access more land as in unfettered land markets. Quantitatively, this distortion captures residual misallocation, over and above that

implied by land insecurity, and helps reconcile the observed land allocations in the data.

IV.C. Village Families

We describe the occupational choices of village families in two steps. First, we characterize the income of each individual j in family i from working in every possible occupation: farm operator, agricultural worker, and nonagricultural worker. Second, we characterize family i 's endogenous allocation of individual members to occupations to maximize total family income.

1. *Farm Operator.* To operate a farm, each family i must assign one individual to be a full-time farm operator to produce the agricultural good and retain their allocated land. The productivity s_{ij} and output distortion τ_{ij} of the family farm are those associated with farm operator j . The operator decides how much land to rent in or out, ℓ_{ij}^{rent} , relative to their endowment $\bar{\ell}$, labor demand, and output supply.

These farm choices, however, are subject to the land insecurity risk that farming families face when renting out land. Specifically, after agricultural production, there is an exogenous probability η that they lose the rented-out land, in which case there is an associated income loss φ_i that is proportional to the amount of rented-out land. The profit maximization problem for family farm i is then given by

$$(3) \quad \pi(s_{ij}, \tau_{ij}) = \max_{n_{ij}^d, \ell_{ij}^{\text{rent}}} \left\{ \tau_{ij} p_a A_a s_{ij} \left[(\bar{\ell} + \ell_{ij}^{\text{rent}})^{\theta} n_{ij}^d{}^{1-\theta} \right]^{\gamma} - q \ell_{ij}^{\text{rent}} - w_a n_{ij}^d + \eta \varphi_i \ell_{ij}^{\text{rent}} \mathbb{1}[\ell_{ij}^{\text{rent}} < 0] \right\},$$

where n_{ij}^d is farm labor input and $\mathbb{1}[\ell_{ij}^{\text{rent}} < 0]$ is an indicator variable that takes the value of one if the family rents out part of their allocated land. Overall, the last term captures the perceived income loss associated with land insecurity for farms that rent out land.

2. *Full-Time Nonagricultural Worker.* An individual j in family i working full-time in the nonagricultural sector, facing labor-mobility barrier ξ_{ij} , earns net income $i^{FN} = w_n h_{ij}(1 - \xi_{ij})$ where the superscript FN denotes “full-time nonagriculture.”

3. *Full-Time Agricultural Worker.* If an individual j , from family i , works as a full-time agricultural worker, they earn wage income. The skills of agricultural workers are homogeneous across all individuals, and normalized to one. We assume that family labor and hired labor are perfect substitutes; hence, all agricultural labor is paid at rate w_a . We denote the income from full-time agricultural work as $i_{ij}^{FA} = w_a$, where the superscript *FA* denotes “full-time agriculture.”

4. *Part-Time Worker.* In the data, workers often earn wages in both agriculture and nonagriculture. To account for this, we allow individuals to work as wage earners part-time in each sector. We denote by n_{ij} the labor supply of individual j from family i to agriculture. The labor supply of the same individual to nonagriculture is $(1 - c - n_{ij})$, where c denotes a fixed cost associated with working part-time. An individual who works part-time in both agriculture and nonagriculture earns income $h_{ij}w_n(1 - \xi_{ij})(1 - c - n_{ij})$ from nonagriculture and $w_a\kappa n_{ij}^\nu$ from agriculture, where κ measures the relative efficiency of labor input in the two sectors. The parameter $\nu < 1$ captures decreasing returns of labor supply to the agricultural sector, allowing for incomplete specialization across sectors for part-time workers (Erosa et al. 2022).

Taking the first-order condition of the total income of a part-time worker with respect to labor supply in agriculture n_{ij} , we obtain the optimal agricultural labor supply

$$n_{ij}^* = \min \left\{ \left(\frac{\nu \kappa w_a}{h_{ij}w_n(1 - \xi_{ij})} \right)^{\frac{1}{1-\nu}}, 1 - c \right\}.$$

The corresponding income is then given by

$$i_{ij}^{PT} = h_{ij}w_n(1 - c - n_{ij}^*)(1 - \xi_{ij}) + w_a\kappa(n_{ij}^*)^\nu,$$

where the superscript *PT* denotes “part-time worker.”

Using the above notation and characterization, we can summarize effective labor supply to the agricultural sector as:

$$(4) \quad n_{ij} = \begin{cases} 1, & \text{if } i_{ij}^{FA} \geq i_{ij}^{FN} \text{ and } i_{ij}^{FA} \geq i_{ij}^{PT}; \\ 0, & \text{if } i_{ij}^{FN} \geq i_{ij}^{FA} \text{ and } i_{ij}^{FN} \geq i_{ij}^{PT}, \\ \kappa(n_{ij}^*)^\nu, & \text{if } i_{ij}^{PT} \geq i_{ij}^{FA} \text{ and } i_{ij}^{PT} \geq i_{ij}^{FN}. \end{cases}$$

5. *Family Allocation Problem.* We characterize how the household allocates its members across occupations. If individual j is chosen to be the farm operator, total income is given by,

$$(5) \quad I_i^a(\text{operator} = j) = \pi(s_{ij}, \tau_{ij}) + \sum_{k \neq j} \max\{i_{ik}^{FN}, i_{ik}^{FA}, i_{ik}^{PT}\}.$$

Operating in agriculture, the household chooses the farm operator that maximizes total income,

$$(6) \quad I_i^a = \max_{j \in J} \{I_i(\text{operator} = j)\}.$$

If the household chooses not to participate in agricultural production, then no family member is assigned to be a farm operator. The household rents out all of their allocated land $\bar{\ell}$, which is then subject to the risk of income loss. In this case, total household income, accounting for land insecurity, is given by

$$(7) \quad I_i^n = \sum_j \max\{i_{ij}^{FN}, i_{ij}^{FA}, i_{ij}^{PT}\} + (1 - \eta)q\bar{\ell} - \eta\phi_i\bar{\ell},$$

where $(1 - \eta)q\bar{\ell}$ is the rental income without reallocation, and $-\eta\phi_i\bar{\ell}$ is the perceived income loss associated with land insecurity. The household chooses to operate a farm if and only if $I_i^a \geq I_i^n$. Hence, total income of village family i is

$$(8) \quad I_i = \max\{I_i^a, I_i^n\}.$$

6. *Role of Land Insecurity.* Insecure property rights over land affect rural household decisions with respect to farming on both the intensive and extensive margins. On the intensive margin (see [equation \(3\)](#)), they discourage farm operators from renting out more land. On the extensive margin (see [equations \(7\) and \(8\)](#)), they deter families from completely abandoning farming, out of fear of losing access to their assigned land. These effects have two critical implications. First, they imply that land insecurity leads to misallocation of land across farms. Due to limited land rentals, the operational scale of farms is more closely tied to a household's administratively allocated use rights. As a result, more productive farmers are constrained in accessing more land. Second, land insecurity acts as a deterrent to migration, resulting in a de facto barrier to labor mobility out of agriculture.

IV.D. *Equilibrium*

We define an indicator variable D_i for family i , with $D_i = 1$ if family i operates a farm in agriculture and $D_i = 0$ otherwise. An individual is one of the following four occupational types: farm operator, full-time agricultural worker, full-time nonagricultural worker, and part-time agricultural and nonagricultural worker. For individual members, we use the following notation to represent optimal occupation choices. We define indicator variables, with $D_{ij}^O = 1$ if individual j is the farm operator and $D_{ij}^O = 0$ otherwise. We denote by $D_{ij}^{FA} = 1$ if individual j is a full-time agricultural worker, $D_{ij}^{FN} = 1$ if a full-time nonagricultural worker, and $D_{ij}^{PT} = 1$ if a part-time worker.

The net rented-in land among households who operate farms is $\int_i D_i \sum_{j \in i} D_{ij}^O (\ell_{ij}^{\text{rent}}) di$, where $D_i D_{ij}^O$ is a product of indicators that household i operates a farm, and individual j in household i is the farm operator. Land rental market clearing requires that

$$(9) \quad \int_i D_i \sum_{j \in i} D_{ij}^O \ell_{ij}^{\text{rent}} di = \int_i (1 - D_i) \bar{\ell} di,$$

where the right-hand term captures the land rented out by households who do not operate farms.

Agricultural labor supply is $\int_i \sum_{j \in i} (1 - D_i D_{ij}^O) n_{ij} di$, where we sum up labor supplied to agriculture for each household member if they are not farm operators. Agricultural labor market clearing then requires

$$(10) \quad \int_i \sum_{j \in i} (1 - D_i D_{ij}^O) n_{ij} di = \int_i D_i \sum_{j \in i} n_{ij}^d D_{ij}^O di.$$

Nonagricultural labor supply consists of two components: efficiency units of labor supplied by urban households $N_n \bar{h}$ and those supplied by rural households. The rural household labor supply to nonagriculture is the sum of that of full-time nonagricultural workers and that from part-time workers,

$$(11) \quad H_n = N_n \bar{h} + \int_i \sum_{j \in i} (1 - D_i D_{ij}^O) \left[(1 - n_{ij} - c) D_{ij}^{PT} + D_{ij}^{FN} \right] h_{ij} di.$$

Denote the agricultural consumption as c_{ai} for rural household i and c_a^n for the representative urban household. We can then

write the agricultural good market clearing condition as,

$$(12) \quad \int_i c_{ai} di + N_n c_a^n = \int_i \sum_{j \in i} D_i D_{ij}^O y_{ij} di,$$

where y_{ij} represents farm output. Similarly, the nonagricultural good market clearing is

$$(13) \quad \int_i c_{ni} di + N_n c_n^n = A_n H_n.$$

Net revenues from the income loss of land rentals, labor mobility barriers, and farm output distortions are rebated back to households uniformly.

With these equilibrium conditions, we define the competitive equilibrium of this economy.

A competitive equilibrium is a set of prices $\{w_a, w_n, p_a, q\}$, allocations of the farm operators $\ell_{ij}^{\text{rent}}, n_{ij}^d, y_{ij}, \pi_{ij}(s_{ij}, \tau_{ij})$; village family incomes I_i , occupation allocations $D_i, D_{ij}^O, D_{ij}^{FA}, D_{ij}^{FN}, D_{ij}^{PT}$, and n_{ij} , consumption of village families (c_{ai}, c_{ni}) , and consumption for the urban households (c_a^u, c_n^u) , such that:

- (i) Given prices, farm operators maximize profits π_{ij} and $\bar{\ell} + \ell_{ij}^{\text{rent}}, n_{ij}^d$ are optimal factor demands.
- (ii) Given prices, nonagricultural firms maximize profits.
- (iii) Given prices, village families maximize income I_i by choosing labor supply n_{ij} from [equation \(4\)](#) and occupations $D_i, D_{ij}^O, D_{ij}^{FA}, D_{ij}^{FN}, D_{ij}^{PT}$ to solve problems [\(6\)](#), [\(7\)](#), and [\(8\)](#).
- (iv) Given prices and income I_i , village households choose consumption (c_{ai}, c_{ni}) to maximize utility in [equation \(1\)](#).
- (v) Given prices, urban households choose consumption (c_a^u, c_n^u) to maximize utility subject to income $w_n \bar{h}$.
- (vi) All markets clear; that is, prices solve [equations \(9\)](#), [\(10\)](#), [\(11\)](#), [\(12\)](#), and [\(13\)](#).

V. ESTIMATION

To estimate the model we use two main data sources: (i) household- and individual-level survey data from the RCRE for 2004–2018, which provides detailed individual labor supply and income data, as well as farm-level inputs and outputs; and (ii) aggregate data from the Chinese Statistical Yearbook supplemented

with revised sectoral employment series from [Brandt and Zhu \(2010\)](#) and [Yao and Zhu \(2021\)](#). We also draw on a supplementary survey undertaken with the RCRE, which covers a more limited number of villages but surveys farmers about their perceptions of land insecurity. A more detailed description of the data and variable definitions are provided in [Online Appendix A](#).

Our estimation strategy involves two steps. First, we parameterize the distributions for sectoral abilities, land market institutions, and labor-mobility frictions of village families. Second, we estimate the parameters of the model to match model moments with empirical moments from the survey and aggregate data. We also discuss our key identification restrictions.

V.A. *Parameterization*

The parametric distributions that we need to specify for individuals in the model are: farm-operating ability, nonagricultural ability, labor-mobility frictions, and residual farm-operating distortions. We also need to specify the distribution for the family-level land insecurity cost.

1. *Ability Distributions.* Each individual j , in family i , draws a farm-operator ability s_{ij} and a nonagricultural ability h_{ij} . We allow for individual abilities to be correlated across members of the same family. In addition, we allow for abilities to be correlated across sectors. In particular, the nonagricultural ability of individual j in family i is given by

$$\log(h_{ij}) = \log(h_i^H) + \log(h_{ij}^I),$$

where h_i^H is a common component for all people in family i and h_{ij}^I is individual j 's idiosyncratic component. Without taking a stance on the source, the common component can capture, for example, the correlation of innate ability or accumulable skills across family members. The agricultural ability of individual j , in family i , is given by

$$\log(s_{ij}) = \log(s_i^H) + \log(s_{ij}^I) + \lambda \log(h_{ij}),$$

where s_i^H is the common family component, s_{ij}^I the individual component, and $\lambda \log(h_{ij})$ the component that is correlated with nonagricultural ability. Note that λ governs the correlation between agricultural and nonagricultural abilities of the same person. We assume that ability components s_i^H , h_i^H , s_i^I , and h_i^I are

all drawn from mean-zero log-normal distributions. The standard deviations of the family components are σ_s, σ_h , respectively, and that of the individual components are $\omega\sigma_s, \omega\sigma_h$, respectively, where ω governs the relative importance of individual components versus family components.

2. Individual-Level Labor-Mobility Frictions. We assume that labor-mobility barriers facing rural individuals working in the nonagricultural sectors are given by,

$$(14) \quad \xi_{ij} = \frac{\exp(\mu_\xi + \varepsilon_{ij}^\xi)}{1 + \exp(\mu_\xi + \varepsilon_{ij}^\xi)}.$$

This parametric form guarantees that labor-mobility barriers are bounded between zero and one, where μ_ξ governs the level of barriers and ε_{ij}^ξ their idiosyncratic component, which we assume follows a normal distribution with mean zero and standard deviation σ_ξ .

3. Residual Idiosyncratic Farm Distortions. Individual j in family i faces implicit distortions to operating a farm, measured by τ_{ij} . We allow for the farm distortion to feature a component that is correlated with farm-operating ability and a component that is uncorrelated, similar to how idiosyncratic wedges are modeled in Bento and Restuccia (2017) and Restuccia (2019),

$$\log(\tau_{ij}) = \zeta \log(s_{ij}) + \varepsilon_{ij}^\tau,$$

where ζ captures the elasticity of distortions with respect to farming ability s_{ij} , and the uncorrelated component ε_{ij}^τ is drawn from a normal distribution with mean zero and standard deviation σ_τ .

4. Family-Level Land Insecurity. In the model, η is the probability that a household loses the use rights to land that it has rented out. Conditional on losing their land, φ_i summarizes the perceived income loss per unit of rented-out land associated with the insecure land rights. We assume that $\log(\varphi_i)$ differs across families and is drawn from a normal distribution with mean μ_φ and standard deviation σ_φ .

V.B. Identification

There are 23 parameters to be determined in our model, including the parameters of the ability distributions and distortions discussed above. We provide a brief overview of our calibration approach, focusing on the elements that are nonstandard. In [Online Appendix B](#) we provide a detailed explanation of how the parameters are assigned values or estimated.

Out of the 23 parameters, 9 are either normalized or set based on a priori information. The remaining 14 parameters are estimated jointly so that model-generated moments for key variables match empirical counterparts. Although the parameters are jointly determined, some moments are more relevant for identifying key parameters. For instance, the distribution of employment across sectors and occupations helps identify the subsistence constraint in agriculture $\bar{a}$, the fixed costs to part-time work in non-agriculture (c), and the relative sectoral efficiency parameters ($\kappa, \bar{h}$).

1. *Ability Distributions.* The moments of the ability distributions are chosen to match empirical moments of dispersion in sectoral incomes (nonagriculture) and farm total factor productivity (TFP) (agriculture). In particular, the dispersion of the family component of nonagricultural ability (σ_h) and the relative importance of the individual components (ω) are identified through the standard deviation of nonagricultural wage income and the rank correlation of nonagricultural wage incomes in the family. The dispersion of farm-operating ability (σ_s) is chosen to match the dispersion of farm TFP in the production data, with the parameter capturing the correlation to nonagricultural ability (λ) identified from the correlation of nonagricultural wage income with farm profit in the family. We discuss in [Online Appendix F](#) the robustness of our results to the sectoral correlation of abilities, governed by λ .

2. *Land Insecurity.* The parameters governing land insecurity are the family-level probability that households renting out land lose the use rights to this land η and the mean μ_φ and standard deviation σ_φ of the family-specific perceived income loss per unit of rented-out land, φ_i .

We determine η directly from data drawn from our supplementary survey. Two pieces of information are critical to its

estimation. First, the observed historical frequency of village-wide land reallocations and land-takings by the state, events that can trigger land loss. Second, the perceived likelihood that a household loses land they have rented out in the course of either a reallocation or land taking. In the case of land expropriation by the state, the risk for a household that rents out their land is that they are not compensated for the land loss. By law, households are to be compensated on the basis of the agricultural value of the land. Up through 2004, the probability that land was affected by either a reallocation or land takings was 10.2%. In addition, the probability a household puts on retaining land they rented out in the event of a land reallocation or land taking was 50%. Our estimate of η is the product of these two probabilities, which is 5.1%. We provide details of our supplementary survey in [Online Appendix A.5](#).

We identify the mean of the family income loss associated with land insecurity μ_φ with the share of village families operating farms in 2004. Our estimation exploits that in the model, the extent to which families participate in agricultural production as operators depends crucially on the extent of land insecurity. In 2004, the share of village families that operate farms was 74%. The dispersion parameter σ_φ governs who these households are. We hence identify the dispersion through its selection implications. If φ_i was the same across families, then families with the highest nonagricultural ability members would choose not to operate farms and move out of agriculture, while families with the lowest nonagricultural ability members would choose to operate farms, implying a counter-factually large wage gap between these two types of families. The dispersion in φ_i across families allows the model to match the wage gap observed in the data between families in agriculture and in nonagriculture. Variation in household size in the data would also be subsumed in the estimated dispersion of φ_i , as we discuss in [Online Appendix E](#).

The income loss associated with land insecurity in the calibrated model is on average around five times the rural household income. In the data, the income loss can be captured by the discounted present value of the land that a household rents out. A back-of-the-envelope calculation suggests that our calibrated value is empirically plausible. With farming the source of one-third to one-half of total family income in 2004, the agricultural revenue from land represents 13%–20% of total household income. Using subjective discount rates of 3%–5%, and a time

horizon of 30 years, the value of land relative to household income is between 2.7 and 6.7, within which our estimate falls.

3. *Labor-Mobility Frictions.* The mean of the labor-mobility frictions to nonagriculture (μ_ξ) is chosen to match the sectoral gap between farming profit and the nonagricultural wage income. The dispersion (σ_ξ) is identified from selection in the labor supply of the part-time workers: with no dispersion in labor-mobility barriers, individuals with higher nonagricultural ability and hence wage income strictly supply more labor to the nonagricultural sector. This sorting pattern is dampened by the dispersion of the labor-mobility barriers. We hence choose σ_ξ to match the observed rank correlation between nonagricultural wage income and labor supply for part-time workers.

4. *Residual Farm Distortions.* The individual-level farm distortion τ_{ij} captures residual farm-operating frictions related to the land institutions that are not directly related to land insecurity. To identify the parameters of the τ_{ij} distribution, we first estimate a summary measure of farm-specific distortions, inferred as a weighted average of actual input deviations from their efficient levels (given farm productivity), what the literature calls revenue productivity or TFPR (Hsieh and Klenow 2009). Then, following Adamopoulos et al. (2022), the correlation of farm distortions with farmer productivity and the dispersion of their nonsystematic component are chosen to match the dispersion of TFPR and, among those who rent in/out land, the correlation of farm TFP and TFPR.

5. *Calibrated Parameters.* Model parameters do not always uniquely determine model-implied moments. Our identification derives from the sensitivity of the model moments to specific parameters as well the direction of the effect. [Online Appendix B](#) outlines in more detail how model parameters map to model moments, with the estimated parameter values reported in [Online Appendix Table B.5](#). The empirical moments of targeted variables, along with the implied model moments are reported in [Table IV](#). Overall, the estimated model matches the data targets well.

6. *Identifying Land Insecurity and Mobility Barriers.* A key objective of our analysis is to identify separately the roles of land

TABLE IV
TARGETED MOMENTS: DATA AND MODEL

Moments	Data	Model
Employment share among village individuals:		
Full-time nonagriculture	0.302	0.299
Part-time	0.207	0.204
Median fraction of part-time hours in agriculture	0.286	0.286
Rank correlation of wages and part-time hours in nonagr.	0.398	0.397
Share of village households with farm operators	0.737	0.736
Sectoral gap: nonagr. wage versus farming profit	0.081	0.081
Family wage differentials:		
Average nonagr. wage, with/without operators	-0.281	-0.282
Wage dispersion among full-time nonagr. workers:		
Standard deviation	0.610	0.608
Within-family correlation	0.558	0.557
Correlation of nonagricultural wage income and farm profit	0.080	0.078
Agricultural production:		
Standard deviation of farm TFP	0.657	0.660
Standard deviation of farm TFPR	0.633	0.631
Rank correlation of farm TFP and TFPR	0.968	0.975
Nominal agr. to nonagr. labor productivity ratio	0.388	0.389

Note. Reporting the 14 moments targeted in the data and the corresponding model estimated moments.

insecurity and labor-mobility barriers. While both frictions lead to overemployment in agriculture and a gap in income between agriculture and nonagriculture, conceptually, the frictions are distinct and empirically work through different margins. The income loss associated with land insecurity is a family-level friction and its effects manifest through the family decision of whether to operate a farm. As a result, land insecurity has a first-order effect on the share of village families operating farms. By contrast, the labor-mobility barrier is an individual-level friction, with its effects manifesting through individual family-member decisions of whether to work in the city. Given that a family needs only one member to operate the farm to secure the land, all other members are not bound by the land-insecurity friction. Hence, labor-mobility barriers are critical for the sectoral income gap between agriculture and nonagriculture. Our identification derives from the sensitivity of the two margins, that is, the share of families operating farms and the sectoral income gap, to land insecurity and labor-mobility barriers.

To appreciate our identification approach, consider the extreme case without land insecurity. Calibrating the labor-mobility

barriers to match the sectoral income gap, as is typically done in the literature, the model misses the share of village families operating farms by more than 30 percentage points (42.4% versus 73.7% in the data). Moreover, the labor-mobility barriers required to account for the sectoral income gap are considerably higher: the labor-mobility barriers are 79.5% on average, compared with 54.0% in the baseline calibration with land insecurity. Our explicit modeling of land rental markets and the associated land insecurity unpacks part of the “black box” of labor-mobility barriers typically emphasized in the literature. Similarly, in the case without labor-mobility barriers and only land insecurity, the model misses the income gap between agriculture and nonagriculture observed in the data. In [Online Appendix Table B.6](#), we more systematically illustrate the sensitivity of the key model moments to changes in the parameters associated with the land-insecurity and labor-mobility frictions. We confirm our identification strategy that the percentage of farm-operating households is particularly sensitive to the level of the land insecurity, while the sectoral income gap is more sensitive to the level of the labor-mobility barriers.

7. Implications for Land Rental Markets. The calibrated model is broadly consistent with the extent of land rentals in the data for China, and the selection of farm households into rentals. In the data, 78% of households who operate farms do not participate in the rental market. While not targeted, in the calibrated model we find that 73% of farm operators neither rent in nor rent out land. The reason for limited participation in rental markets in the model is that with land insecurity, renting out land is associated with an expropriation risk which effectively reduces the return on land that is rented out. Our model endogenously generates the bulk of limited participation in the rental market without resorting to exogenous fixed costs or wedges between reservation prices to rent in and rent out. On the intensive margin of rentals, among those who rent in land, our model generates an average rental of 0.98 of average farm size, similar to that in the data (0.89). Our model is also consistent with the data regarding who rents in land. In the data, the correlation between farm productivity and an indicator for renting in land is 0.06, a sign of misallocation, which could occur when rental is based on familial relationships rather than productivity. Without targeting it in the calibration, our model also implies a weak correlation of 0.27.

In [Online Appendix E](#) we also show that if a household member operating the family farm leaves the sample (e.g., passes away), they are replaced with a family member who was previously working in nonagriculture. This is consistent with the model implications that families assign as operators members that do not necessarily have a comparative advantage in agriculture.

VI. QUANTITATIVE ANALYSIS

We evaluate the quantitative role of land security and labor-mobility frictions in 2004, the year of our baseline estimation, and then analyze their evolution over time. We also study the link between land insecurity and misallocation in agriculture.

VI.A. *Land Security and Labor-Mobility Barriers*

To assess the importance of land insecurity and labor-mobility barriers on outcomes, we perform two quantitative experiments from the benchmark economy. First, we assess the role of land insecurity by setting $\eta\varphi_i = 0$ for all families. Second, we eliminate labor-mobility barriers by setting ξ_{ij} to zero for all individuals and families. We examine the quantitative implications of these changes for sector employment, agricultural labor productivity, and other aggregate outcomes.

The results of these counterfactual experiments are presented in [Table V](#). Eliminating land insecurity ($\eta\varphi_i = 0$) in the second column results in a substantial decline in the fraction of households who operate farms from 73.6% to only 28.2%. This suggests that more than half of farms in the baseline are “zombie” farms, that is, low-productivity rural households that operate farms simply to avoid losing their allocated land. As a result, average farm size more than doubles with the exit of “zombie” farms. The share of employment in agriculture among village households falls less precipitously from 56.4% to 45.6%.

There is improved selection into farming, both within families as well as across families. The ability of the median operator rises by 22.1% as the percentage of farming households for whom the farm operator has the highest agricultural ability increases from 61.2% in the baseline to 69.6%. Overall, agricultural labor productivity rises by 24.7%, the bulk of which is due to the smaller yet more selected set of farmers, as we show in

TABLE V
THE ROLE OF LAND SECURITY AND LABOR-MOBILITY BARRIERS

	Baseline	Land security	No labor barriers
Village families operating farms (%)	73.6	28.2	49.1
Agricultural employment share among villagers (%)	56.4	45.6	46.3
Δ Agricultural output (%)	—	+0.9	−4.9
Δ Agricultural labor productivity (%)	—	+24.7	+15.9
Δ Median log farm-operator ability (%)	—	+22.1	+3.9
Δ Nonagricultural output (%)	—	+4.0	+4.6
Δ Real GDP per capita (%)	—	+3.5	+2.7
Within-household selection in farming:			
% of farm operators with highest s_{ij}	61.2	69.6	62.6
Nominal agricultural productivity gap	2.57	2.20	1.39

Notes. Statistics for the baseline model, counterfactual land security ($\eta\varphi_i = 0$ for all i), and counterfactual no labor-mobility barriers ($\xi_{ij} = 0$). The first two and the last two statistics are in levels, while all other statistics are displayed as differences compared to the baseline model.

[Online Appendix H](#). With labor released from agriculture to non-agriculture, aggregate nonagricultural output increases by 4.0%. The total increase in real GDP per capita, calculated using a chain-type quantity index, is 3.5%.

Eliminating labor-mobility barriers ($\xi_{ij} = 0$) in the third column of [Table V](#) has similar effects on key moments in the data, however several differences emerge. First, removal of these constraints results in a smaller reduction in the share of households operating farms, but a similar reduction in agricultural employment. This confirms our identification strategy that land insecurity matters especially at the household level, while labor-mobility barriers matter more at the individual level. Accompanying these changes are smaller improvements in farm-operator ability through selection, which only rises by 3.9%. Second, there are much larger reductions in output in farming that are partially offset by larger gains in nonagricultural output. Real GDP per capita rises by 2.7% compared with 3.5% when we eliminate land insecurity. Overall, labor-mobility barriers have similar effects as land insecurity on the share of employment in agriculture and GDP per capita but weaker effects on selection and agricultural labor productivity.

We use our framework to shed light on the nominal agricultural productivity gap (APG), defined as the ratio in value added per unit of labor input between agriculture and nonagriculture. It

is well documented that poor countries tend to have a large APG (Gollin, Lagakos, and Waugh 2014b). Our quantitative framework suggests that both land insecurity and labor-mobility barriers contribute to the size of the APG. In our baseline economy, the targeted APG is 2.57-fold, but this drops to 2.20 when we eliminate land insecurity and to 1.39 when we eliminate labor-mobility barriers. Eliminating labor-mobility barriers has a larger effect on the APG despite a smaller effect on agricultural labor productivity. This highlights that closing the nominal labor productivity gap between sectors does not necessarily close cross-country agricultural productivity differences (Restuccia, Yang, and Zhu 2008; Gollin, Lagakos, and Waugh 2014a). In Online Appendix D, we examine the role of within-family selection for real agricultural productivity and the nominal agricultural productivity gap.

VI.B. *Evolution of Frictions over Time*

We leverage the survey data through 2018 to assess the evolution of land insecurity and labor-mobility barriers over time. Our baseline model is calibrated to 2004 moments.

We recalibrate the model to match data moments in 2018, which allows us to back out the land insecurity ($\eta\varphi_i$) and labor-mobility (ξ_{ij}) parameters at the end of our period. Details regarding the recalibration in 2018 are provided in Online Appendix C. We use the estimated values of land insecurity and labor-mobility barriers to perform two separate counterfactual experiments in our baseline economy. First, we change the land-insecurity parameters to match their estimated values for 2018. To do this we: (i) reduce the probability that a household renting out their land loses it through expropriation, η , from 5.1% in 2004 to 0.8% in 2018; and (ii) reduce the expected income loss associated with land insecurity ($\eta\varphi_i$) to match the 2018 ratio of 5.8% of annual family income (versus 27.9% in 2004). Second, maintaining land-insecurity parameters at their 2004 levels, we change labor-mobility barriers to their 2018 estimated values.

The effects of these experiments are reported in Table VI. When we change the extent of land insecurity alone (second column) from their 2004 to their 2018 level, the percentage of rural households who operate farms falls substantially from 73.6% to 37.6%. As a result, agricultural labor productivity increases by 20.6%. This result suggests that land insecurity in 2018 is much less severe compared with 2004, consistent with the measured

TABLE VI
THE EVOLUTION OF FRICTIONS OVER TIME

	Baseline	2018 land security	2018 labor barriers
Village families operating farms (%)	73.6	37.6	83.5
Agricultural employment share among villagers (%)	56.4	47.3	61.8
Δ Agricultural output (%)	—	+1.1	+3.3
Δ Agricultural labor productivity (%)	—	+20.6	−5.7
Δ Median log farm-operator ability (%)	—	+14.6	−0.4
Δ Nonagricultural output (%)	—	+3.4	−2.3
Δ Real GDP per capita (%)	—	+3.1	−1.5
Within-household selection in farming: % of farm operators with highest s_{ij}	61.2	66.6	63.4
Nominal agricultural productivity gap	2.57	2.32	3.41

Notes. Statistics for the baseline 2004 economy and two counterfactuals with frictions at their 2018 value. We replace the level of land frictions η and μ_φ such that the average $\left(\frac{\eta\varphi_i\bar{\ell}}{T_i}\right)$ matches that of the 2018 level (second column) and labor-mobility frictions ξ_{ij} to 2018 values (third column). The first two and the last two statistics are in levels, while all other statistics are displayed as differences compared to the baseline 2004 economy.

decline in the frequency of village-wide land reallocations and with a recent literature emphasizing the role of institutional reforms in China in greatly alleviating frictions in the land market.

Over the period between 2004 and 2018, entire families in the RCRE survey occasionally migrate to urban areas and hence are replaced in the sample. If these households dropped from the sample are less likely to be engaged in farming before migration than newly added households, our calculation could underestimate the decline in the percentage of households who operate farms and in turn underestimate the improvement in land security over time. We examine the robustness of our estimates by also considering a balanced sample of households between 2004 and 2018 and find that our estimates do not differ substantially and that focusing on the unbalanced panel delivers more conservative estimates for the drop in land insecurity. We provide more details and a discussion of attrition in our sample in [Online Appendix A.6](#).

When we change labor-mobility barriers alone from 2004 to their 2018 values (third column), we find that agricultural employment actually increases and agricultural labor productivity declines slightly. This suggests that labor-mobility barriers between agriculture and the nonagricultural sectors have increased over our sample period.

The comparison in the evolution of land and labor frictions has important implications for our understanding of migration costs. Simply looking at data on migration flows, we might

TABLE VII
INTERACTION BETWEEN LAND SECURITY AND MISALLOCATION WITHIN
AGRICULTURE

	Baseline	Land security	No idio. farm dist.	+ Land security
Village families operating farms (%)	73.6	28.2	56.5	2.2
Agr. employment share among villagers (%)	56.4	45.6	36.9	20.2
Δ Agricultural output (%)	—	+0.9	+56.9	+87.7
Δ Agricultural labor productivity (%)	—	+24.7	+140.0	+424.5
Δ Median log farm-operator ability (%)	—	+22.1	−1.9	+161.3
Δ Nonagricultural output (%)	—	+4.0	+6.0	+10.6
Δ Real GDP per capita (%)	—	+3.5	+10.6	+17.2
Within-household selection in farming:				
% of farm operators with highest s_{ij}	61.2	69.6	52.3	93.2
Nominal agricultural productivity gap	2.57	2.20	5.70	3.75

Notes. Statistics of the baseline model for three counterfactual experiments: land security ($\eta\varphi_i = 0$), no idiosyncratic farm distortions ($\tau_{ij} = 1$), and both. Statistics for deviations Δ are from the baseline model.

infer on the basis of the reallocation of labor from agriculture to nonagriculture over time that migration costs have fallen. But looking at the distinct components of migration costs, we find that during our sample period, the reduction in overall migration costs is due exclusively to a reduction in land insecurity rather than labor-mobility barriers, the focus of a large literature. In fact, our estimates suggest that residual labor-mobility barriers have increased slightly over time.

VI.C. Land Security and Misallocation within Agriculture

We examine the effect of reforms in the agricultural sector that simultaneously improve land security and eliminate the idiosyncratic farm distortions in agriculture. Recall that idiosyncratic farm distortions τ_{ij} capture residual misallocation in agriculture beyond that captured by land insecurity.

To examine the interaction of land insecurity with idiosyncratic farm distortions, we implement two counterfactual experiments. First, in the baseline model we remove all idiosyncratic farm distortions by setting $\tau_{ij} = 1$ for all individuals and households. Second, we remove land insecurity ($\eta\varphi_i = 0$ for all families). Results of these experiments are reported in Table VII, third and fourth columns, along with effects of the earlier land-security experiment, reproduced in the second column. Since in practice

agricultural reform is unlikely to eliminate entirely both land insecurity and the farm-level distortions, our results capture an upper bound of these effects.

Consistent with the findings in Adamopoulos et al. (2022), removing idiosyncratic distortions has a large positive effect on agricultural productivity. The striking finding is that land security has much bigger productivity effects in the economy without idiosyncratic farm distortions ($\tau_{ij} = 1$). Comparing the third and fourth columns, implementing land security in an economy without idiosyncratic farm distortions increases agricultural labor productivity by three times, while implementing land security alone increases agricultural labor productivity by 25% in the baseline economy with idiosyncratic distortions. This result illustrates an important complementarity between these frictions.

The reason for the substantial quantitative interaction between land security and residual idiosyncratic farm distortions is twofold. First, on the extensive margin, in an economy without idiosyncratic farm distortions land security encourages the exit of less productive farmers, resulting in a substantial increase of median log farm-operator ability of 161.3%. By contrast, in the presence of residual misallocation in agriculture, land security does not feature as strong selection in farming as implied by the more modest increase in median log farm-operator ability (22.1%). Second, on the intensive margin, land security without idiosyncratic farm distortions enables the reallocation of land from less to more productive farms, whereas with idiosyncratic distortions the land reallocation does not necessarily flow to the most productive farms. Also note that eliminating idiosyncratic farm distortions in the presence of land insecurity aggravates the misallocation of farming talent, with only 52.3% of the farming households having the most productive operator. Agricultural reforms that improve land security and resource allocation in agriculture can attain a substantial increase in agricultural productivity due in part to improved selection in households.

VII. EXTENSIONS

We extend the main framework to incorporate a rural nonagricultural sector and then age differences among family members. We also examine regional heterogeneity by comparing peri-urban and remote villages.

TABLE VIII
STATUS AND LOCATION OF NONAGRICULTURAL WORK BY RURAL INDIVIDUALS

Variable	2004	2009	2013	2018
Share full-time nonagriculture (%)	30.2	38.0	43.3	55.8
Rural (%)	14.8	17.1	20.6	29.6
Urban (%)	15.4	21.0	22.7	26.2
Share part-time nonagriculture (%)	20.7	17.9	16.1	12.0
Rural (%)	15.9	13.7	12.3	9.0
Urban (%)	4.8	4.2	3.7	3.0

Notes. All statistics are calculated from micro-level data from the RCRE survey, as shares out of the total numbers of employed individuals in the survey. For each year the shares of rural and urban workers sum up to the total share in nonagriculture by full- and part-time status.

VII.A. Rural and Urban Nonagricultural Sectors

A key feature of China's structural transformation is the significant role played by nonagricultural employment opportunities in rural areas. In Table VIII, we document the changes between 2004 and 2018 in the level and composition of nonagricultural employment among rural households, where we distinguish between individuals working in urban and rural locations. More than half of those leaving agriculture find work in rural nonagriculture, with the percentage slightly higher earlier in our sample period.

To capture this feature, we extend our model to allow for two nonagricultural sectors: a rural nonagricultural sector (r) in rural townships and an urban nonagricultural sector (u) in urban centers. With this extension, the model can distinguish between spatial reallocation (rural versus urban areas) and sectoral reallocation (agriculture versus nonagriculture).

The production of the two nonagricultural sectors are summarized by constant returns to scale technologies,

$$Y_r = A_r H_r, \quad Y_u = A_u H_u,$$

where A_r and A_u are the productivity parameters and H_r and H_u are the total labor inputs in efficiency units. The outputs from the two nonagricultural sectors are perfect substitutes in consumption. Individuals in villages can choose to work locally in the rural nonagricultural sector or migrate to the cities to work in the urban nonagricultural sector. Work in the two nonagricultural sectors is subject to potentially different individual-level labor mobility barriers, given by ξ_{ij}^r and ξ_{ij}^u for the rural and urban nonagricultural sectors, respectively. An individual choosing full-time

nonagriculture employment works in the urban non-agricultural sector if the wage rate net of mobility barriers exceeds that in the rural nonagricultural sector. People can also choose to work part-time in each nonagricultural sector, with c_r (c_u) capturing the fixed cost associated with part-time farming and rural (urban) nonagricultural work. Note that we do not allow for part-time work between rural and urban nonagricultural sectors in the model as this is extremely rare in the data.

We parameterize the two labor-mobility barriers ξ_{ij}^r and ξ_{ij}^u as in [equation \(14\)](#), with means given by μ_r and μ_u and the same dispersion σ_ξ . Extending our model with a rural nonagricultural sector introduces three more parameters to determine (A_r, μ_r, c_r) , in addition to (A_u, μ_u, c_u) . We normalize $A_r = 1$ and calibrate relative urban productivity A_u to match the split of full-time non-agricultural employment between the rural and urban sectors. We restrict μ_r (μ_u) and c_r (c_u) to match the sectoral income gap between rural (urban) nonagricultural wages and farming profit; and the shares of part-time workers of the rural (urban) nonagricultural sectors. The estimated parameter values are provided in [Online Appendix Table C.8](#).

We find that it is less costly to work part-time in rural nonagriculture than in urban nonagriculture ($c_r = 0.06$ versus $c_u = 0.11$) as rural nonagriculture does not require individuals to move. At the same time, we estimate that labor-mobility barriers are higher on average for the rural nonagricultural sector ($\mu_r = 1.01$) than for the urban nonagricultural sector ($\mu_u = 0.42$). Although the wage gap between rural nonagriculture and agriculture is larger than that between urban nonagriculture and agriculture, individuals are less likely to move to full-time rural nonagricultural work. The model rationalizes this behavior with a larger mobility barrier to rural nonagriculture.

We perform the same experiments as in our main model, eliminating in turn land insecurity ($\eta\varphi_i = 0$ for all i) and labor-mobility barriers ($\xi_{ij}^r = \xi_{ij}^u = 0$). The results of these experiments are summarized in [Table IX](#), Panel A. Paralleling our baseline results in [Table V](#), we find that both experiments have a similar effect on the share of labor in agriculture, but other differences emerge. Land security has a larger effect on the fraction of village families who operate farms, selection, and agricultural productivity, while eliminating labor-mobility barriers results in larger gains in nonagricultural output.

TABLE IX
EXTENSIONS

	Baseline	Land security	No labor barriers
Panel A: Rural and Urban Nonagriculture			
Village families operating farms (%)	73.6	27.7	47.4
Agricultural employment share among villagers (%)	55.3	46.6	46.1
Δ Agricultural output (%)	—	-0.2	-5.6
Δ Agricultural labor productivity (%)	—	+18.4	+13.1
Δ Median log farm-operator ability (%)	—	+16.6	+0.4
Δ Nonagricultural output (%)	—	+3.5	+9.8
Δ Real GDP per capita (%)	—	+2.9	+6.6
Panel B: Age Differences			
Village families operating farms (%)	73.6	31.9	44.7
Agricultural employment share among villagers (%)	56.7	46.1	43.4
Δ Agricultural output (%)	—	+1.1	-5.6
Δ Agricultural labor productivity (%)	—	+24.4	+23.3
Δ Median log farm-operator ability (%)	—	+19.6	+5.9
Δ Nonagricultural output (%)	—	+4.0	+10.3
Δ Real GDP per capita (%)	—	+3.6	+6.8

Note. Statistics for the baseline estimated model versus our two counterfactual experiments, land security ($\eta\varphi_i = 0$ for all i) and no labor-mobility barriers ($\xi_{ij}^L = \xi_{ij}^H = 0$). Panel A shows results for the extended model with rural and urban nonagriculture, and Panel B for the extended model with rural and urban nonagriculture and young-old age differences. The percentage of village households operating farms and the sectoral employment shares among villagers are in levels, while all other statistics are displayed as differences compared to the baseline model.

VII.B. Age Differences within Family Members

Several recent studies have documented the phenomenon of structural transformation by age (Hobijn, Schoellman, and Vindas Quesada 2018; Porzio, Rossi, and Santangelo 2022). In the Chinese context, we also find that older individuals are substantially overrepresented in agriculture, either as farm operators or as full-time agricultural workers, see Online Appendix Tables A.1 and A.2. We further extend our framework to include age differences, in addition to the two nonagricultural sectors described above. This extension allows an examination of the effect of age differences for family, occupational, and sectoral selection and their aggregate implications for productivity and structural change. We perform the same counterfactuals of improving land security and removing labor-mobility barriers and use the framework to disentangle the sources of the overrepresentation of old adults in agriculture.

We incorporate age demographics in the model by assuming that each individual j in family i can be either old with probability p_o , or young with complementary probability $1 - p_o$. We allow individual-level ability distributions and labor-mobility barriers to differ by age group. Specifically, the individual components of nonagricultural and farm-operating abilities, h_{ij}^I and s_{ij}^I , are mean zero for old adults (as in the baseline), but have means μ_h^y and μ_s^y for the young adults. Rural and urban labor-mobility barriers of the young and old have the same parametric form as in the baseline [equation \(14\)](#). The means of the labor-mobility barriers to rural and urban nonagriculture are μ_r and μ_u for the young adults, but are shifted by μ_r^o and μ_u^o for the old adults.

In the data, we define young individuals as those between 16 and 44 years of age, and old as those age 45 and older, but our results are robust to alternative reasonable categorizations of age groups. Adding age differences to our framework introduces four more parameters to the estimation $\{\mu_h^y, \mu_s^y, \mu_r^o, \mu_u^o\}$. We use four additional moments in the data to restrict these parameters: the young-old income differentials among rural nonagricultural workers and among farm operators, and the fraction of individuals who are old among full-time rural nonagricultural workers and among full-time agricultural workers or farm operators. All calibrated parameter values are provided in [Online Appendix Table C.8](#).

Compared with old adults, we find that the young have an absolute advantage in both sectors but a comparative advantage in nonagriculture ($\mu_h^y = 0.20$ and $\mu_s^y = 0.15$). In addition, the estimated labor-mobility barriers for the old are higher, especially to the urban nonagricultural sector ($\mu_r^o = 0.41$ and $\mu_u^o = 1.10$), which implies that the old are particularly less mobile across space. The relatively lower labor-mobility barriers in the urban nonagricultural sector for the young could reflect nonpecuniary benefits for young individuals in the cities, such as life cycle growth, urban amenities, or social welfare.

We perform the same experiments of providing land security ($\eta\varphi_i = 0$ for all i) and eliminating labor-mobility barriers ($\xi_{ij}^r = \xi_{ij}^u = 0$), which we summarize in [Table IX](#), Panel B. The results are similar to our baseline results in [Table V](#) and those reported with the two nonagricultural sectors in [Table IX](#), Panel A. Improving land security has a stronger effect on reducing the fraction of village families who operate farms and improving selection into farming. However, now those that select into

TABLE X
OCCUPATION DIFFERENCES BY AGE GROUPS IN THE MODEL

	Baseline	Land security	No labor barriers	Both
Old				
Agricultural employment share (%)	66.3	51.7	37.7	30.8
Operator	31.3	13.2	15.4	8.7
Worker	35.0	38.5	22.3	22.1
Young				
Agricultural employment share (%)	38.6	27.5	30.0	24.4
Operator	19.9	8.9	14.6	9.1
Worker	18.6	18.6	15.3	15.3

Notes. The share of old and young adults among village households in agricultural employment (and between operator and agricultural worker) for the baseline model and counterfactuals.

operating farms are disproportionately from the young group. Eliminating the labor-mobility barriers has a stronger effect on aggregate output per capita.

We use our framework to shed light on the fact that the old are much more likely to be involved in farming. Two forces are at work. First, old adults have a comparative advantage in agriculture. Second, labor-mobility barriers are more severe for the old than for the young. Both factors predispose families to choose older family members as farm operators.

Table X reports separately agricultural employment shares for young and old individuals and the split between operators versus agricultural workers, for the (extended) baseline and our counterfactuals. In our baseline, 66.3% of old individuals work exclusively in agriculture, either as farm operators or full-time agricultural workers, compared with 38.6% of young individuals. Eliminating land insecurity, agricultural employment shares decline among both young and old, with a slightly larger decline among the old. Eliminating labor-mobility barriers, the agricultural employment shares also decline substantially among both young and old groups, but especially among the old. This is consistent with our finding that labor-mobility barriers are more severe for the old, and key in explaining the young-old gap in agricultural employment.

In the last column of Table X, we eliminate both land insecurity and labor-mobility barriers. In this case, a gap in agricultural employment between young and old of 6.3 percentage points still exists. Out of a total gap of 27.8 percentage points

TABLE XI
MODEL WITH REGIONAL HETEROGENEITY

	Baseline	Land security	No labor barriers
Panel A: Peri-urban			
Village households operating farms (%)	57.2	27.8	33.2
Agricultural employment share among villagers (%)	48.2	40.9	37.0
Δ Agricultural output (%)	—	+1.3	−6.7
Δ Agricultural labor productivity (%)	—	+19.2	+21.4
Δ Median log farm-operator ability (%)	—	+21.1	+9.4
Δ Nonagricultural output (%)	—	+2.6	+6.8
Δ Real GDP per capita (%)	—	+2.4	+4.3
Panel B: Remote			
Village households operating farms (%)	77.8	36.4	52.2
Agricultural employment share among villagers (%)	60.0	51.5	48.8
Δ Agricultural output (%)	—	+0.4	−5.6
Δ Agricultural labor productivity (%)	—	+17.1	+16.2
Δ Median log farm-operator ability (%)	—	+14.2	+2.5
Δ Nonagricultural output (%)	—	+3.6	+9.1
Δ Real GDP per capita (%)	—	+3.1	+5.6

Note. Statistics for the baseline estimated model versus our two counterfactual experiments, land security ($\eta\psi_i = 0$ for all i) and no labor-mobility barriers ($\xi_{ij}^l = \xi_{ij}^u = 0$). Panel A shows results for the peri-urban area and Panel B shows results for the remote area. The percentage of village households operating farms and the sectoral employment shares among villagers are in levels, while all other statistics are displayed as differences compared to the baseline model.

in the calibrated model (and in the data), 20.1 percentage points are the product of labor-mobility barriers and 3.7 percentage points due to land insecurity, hence land insecurity accounts for only about 15% of the age gap in agricultural employment. However, land insecurity contributes about 40% to the age gap among farm operators. This finding is consistent with the idea that in the face of land insecurity, a family’s choice of farm operator may not be based on comparative advantage. Rather, a family could choose a person with few outside options (and thus low opportunity cost) to operate the family farm. In our estimation, this typically means choosing an old individual to be the operator.

VII.C. Village Heterogeneity

Rich heterogeneity exists across villages in China. We assess how land security and labor-mobility barriers differ across one dimension of village heterogeneity: proximity to urban centers. On the basis of a village’s location indicator provided in the survey,

we divide villages into two groups: peri-urban and remote. For 2004, this classification implies that 37% of farm household observations are from peri-urban villages and the remaining from remote villages. Ex ante, we expect households in the peri-urban areas to have better access to off-farm opportunities, and thus face lower labor-mobility barriers. Land expropriations by the state for nonagricultural purposes are more prominent in peri-urban areas. However, officials in these villages are under more careful supervision by higher-level authorities, which may make it easier for village households to assert their rights to compensation in the event of a land taking by the state. In [Online Appendix A.5](#) we calculate the probability that households renting out land lose the use rights (η) separately for peri-urban and remote villages, and we find that overall the peri-urban areas have slightly lower η .

For each region, we recalibrate the model. The aggregate target moments remain the same, while micro-data moments are calculated using observations from each set of villages. In each recalibration, we assume that all villages in the economy are either only peri-urban or remote. The calibrated parameter values are reported in [Online Appendix Table C.8](#). Consistent with our priors, peri-urban villages feature a substantially lower fraction of households operating farms and a lower share of people employed in agriculture. Either off-farm opportunities are more appealing to people in the peri-urban region, or land insecurity and mobility frictions are less severe. We now take a deeper look at land security and labor-mobility barriers for peri-urban and remote regions.

Land insecurity is substantially less severe in the peri-urban than in remote villages. Land income loss relative to annual family income is 45% lower in peri-urban compared to remote regions. As shown in [Table XI](#), providing land security ($\eta\varphi_i = 0$) in the peri-urban area reduces the percentage of village households operating farms by 19.4 percentage points. As expected, the effects are much larger in the remote villages—the percentage of village households operating farms drops by 41.4 percentage points.

The comparison of labor-mobility barriers between the two types of villages is slightly more complicated. Labor-mobility barriers to the rural nonagricultural sector are substantially lower in peri-urban compared with remote villages. This is reflected in the estimated parameters $\mu_r = 0.48$ and $\mu_r^o = 0.36$ for peri-urban compared with $\mu_r = 0.90$ and $\mu_r^o = 0.43$ for remote villages. Labor-mobility barriers to the urban nonagricultural sector are

similar between peri-urban and remote areas. Removing all labor-mobility barriers has slightly larger effects in remote villages in terms of nonagricultural output and real GDP per capita, but similar effects for the percentage of village households operating farms and the agricultural employment share among village members.

A distinct feature of China's economic growth is the prominence of township and village enterprises. These enterprises are often located in towns and peri-urban areas, providing off-farm opportunities to village families (Brandt and Zhu 2010). Since these enterprises are part of the rural nonagricultural sector in our model, it is unsurprising that labor-mobility barriers to the local nonagricultural sector are lower for village households living in peri-urban areas. The fact that peri-urban and remote village members face similar barriers to the urban nonagricultural sector indicates that the barriers may largely arise from factors other than geographical distances.

There are several reasons land frictions might be lower in the peri-urban area. First, agriculture is less important in these localities. With more off-farm opportunities and lower labor-mobility barriers, there is a smaller share of households and people that are farming. This translates into a lower demand for farmland and less lobbying pressure by village households on village leaders to reallocate land. Second, because peri-urban villages are nearer to cities/towns, village leaders in these localities may be under more careful scrutiny of local government, resulting in less uncompensated expropriation.

VIII. CONCLUSIONS

We have developed a unified framework to study jointly the effect of land insecurity and other labor-mobility barriers on agricultural productivity, structural change, and economic development. A key ingredient of our model is that we allow for selection of individuals in families, across sectors, and across space. This framework along with the rich heterogeneity in the panel data for China allow us to quantify the role and impact of land insecurity and other labor-mobility frictions over time.

We find that in 2004, effective migration costs are high, with land insecurity and all other labor-mobility barriers playing an equal role in these costs. By 2018, overall migration costs have dropped, with all of the decline accounted for by improved land security. In fact, labor-mobility costs unrelated to land

institutions increased slightly between 2004 and 2018. This implies that attributing the substantial increase in migration we observe to a drop in labor-mobility barriers is misleading. We also find that farm distortions in agriculture, interacting with selection, amplify the positive effect of land security on productivity.

Our findings have important implications for development policy. Weak property rights over rural land deter land rentals across rural households and migration out of agriculture. Since 2003, there have been significant reforms in China that have strengthened household property rights in land. Our results are consistent with land policy having been central to China's structural change and agricultural productivity growth over the past two decades. While there have been reforms since the early 2000s to migration policy, particularly the *hukou* registration system, these changes have not had an effect on out-migration from agriculture, structural change, and growth. The implication is that there are effective labor-mobility barriers facing rural households that remain high. This could be due to a nonuniform reform or implementation of the migration policy (e.g., it is actually more difficult to move to the coastal megacities), or due to other factors that deter living in the cities, such as higher housing prices. Our results indicate that there is scope for further agricultural productivity growth and structural change from reductions in other direct and indirect labor-mobility barriers. As a result, future policy reforms that reduce effective rural-urban migration costs can have large welfare benefits.

We have focused on the interaction between land insecurity and migration decisions. Many other factors also affect structural transformation and rural-urban migration and could have interesting interactions with insecure land tenure, including international trade (Tombe and Zhu 2019), capital-labor substitution (Chen 2020a, 2020b), choices between food crops and cash crops (Adamopoulos and Restuccia 2020), and local entry barriers of establishments (Brandt, Kambourov, and Storesletten 2020). We leave the study of the interaction of these factors with insecure land tenure for future research.

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SUPPLEMENTARY MATERIAL

An [Online Appendix](#) for this article can be found at *The Quarterly Journal of Economics* online.

DATA AVAILABILITY

The data underlying this article are available in the Harvard Dataverse, <https://doi.org/10.7910/DVN/G3WBU7> (Adamopoulos et al. 2024).

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